



Tradeasia
Academy 9.0
Global Marketplace Academy

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ORDER MANAGEMENT SYSTEM MODULE

Group 2 – 10th June 2026



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CHAPTER 1:

EXECUTION GAP ANALYSIS

Execution Gap Analysis



1. RFQ SYSTEM EXISTS

TradeChem already provides an RFQ system that allows buyers to request quotations from suppliers based on specific product requirements.



2. COMMUNICATION & NEGOTIATION

Buyers and suppliers can communicate and negotiate key terms such as pricing, quantity, delivery conditions, and payment arrangements.



3. EXECUTION GAP

Once an RFQ is accepted, there is no centralized system to manage execution. Orders lack:

- **Visibility**
- **Accountability**
- **Status Tracking**
- **Document Management**
- **Structured Coordination among stakeholders**



4. OMS SOLUTION REQUIRED

An Order Management System (OMS) is needed to transform an accepted quotation into a fully managed order by:

- **Tracking progress**
- **Assigning responsibilities**
- **Coordinating logistics**
- **Managing documents**
- **Handling exceptions**
- **Ensuring successful completion**



CHAPTER 2: OMS OBJECTIVES



OMS OBJECTIVES

Objectives

Convert RFQ into an Order

Lock Commercial Terms

Assign Stakeholders

Track Execution

Impact

Transform accepted RFQs into formal orders with unique Order IDs and structured workflows.

Secure agreed pricing, quantity, Incoterms, delivery schedules, and payment terms.

Define responsibilities for buyers, suppliers, logistics partners, agents, and administrators.

Monitor order progress through real-time status updates from confirmation to delivery.



OMS OBJECTIVES

Objectives

Impact

Manage Documents

Store, validate, and track critical trade documents throughout the order lifecycle.

Coordinate Logistics

Enable seamless shipment planning, tracking, and delivery among all parties.

Handle Exceptions

Manage delays, disputes, compliance issues, and operational disruptions efficiently.

Close the Order

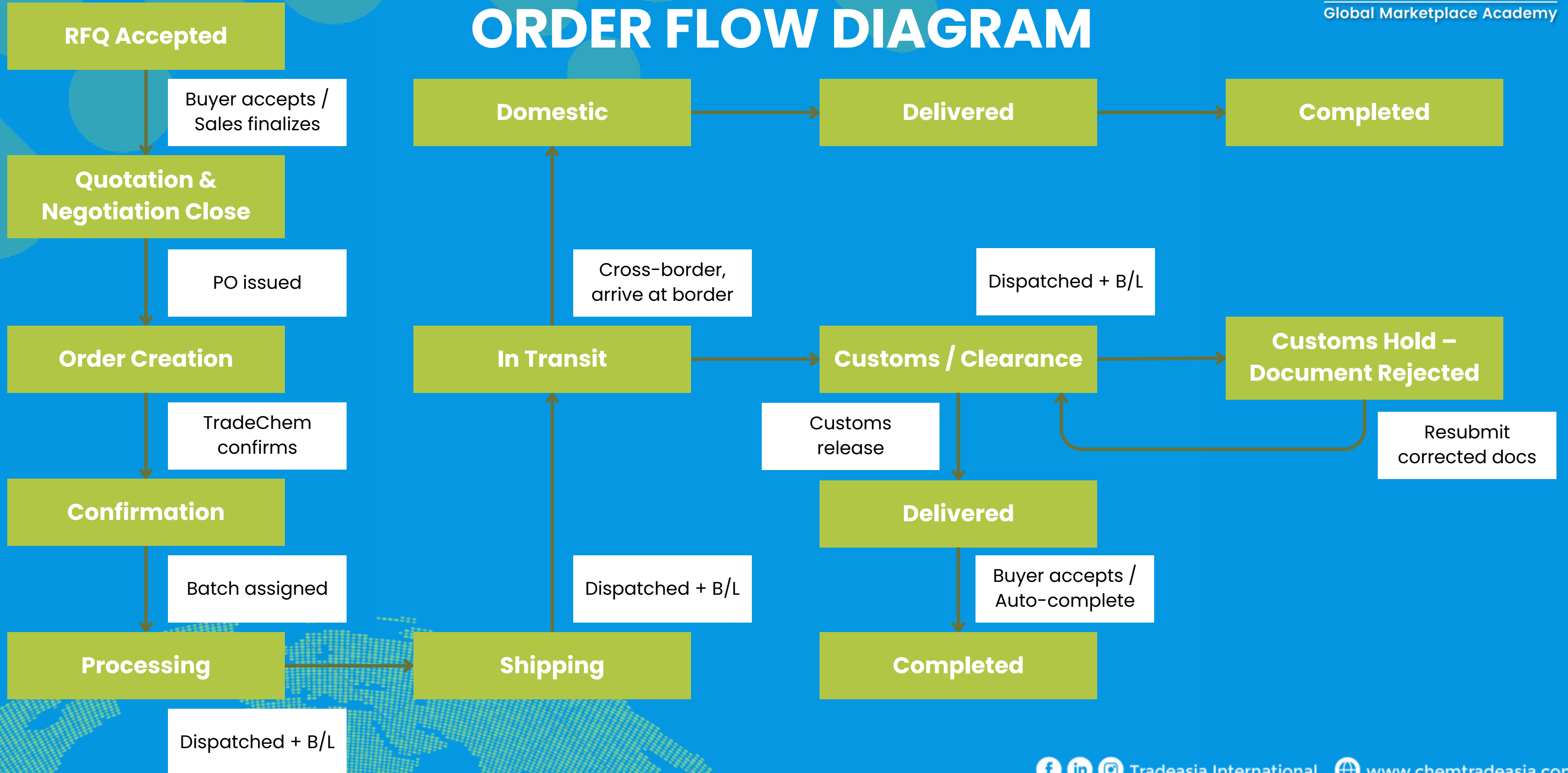
Verify completion, finalize documentation, release payments, and archive records.



CHAPTER 3:

ORDER FLOW DIAGRAM

ORDER FLOW DIAGRAM





CHAPTER 4:

STATUS DEFINITION TABLE

STATUS, RESPONSIBILITY, ACTION & NOTIFICATION

Status	Responsible Party	Required Action (key)
RFQ Accepted	Buyer	Store RFQ PDF + preliminary quote; validate UN number if dangerous goods; upload statement of intended use.
Quotation & Negotiation Close	Agent (Sales Rep) + Buyer	Lock price, Incoterms, payment terms, tolerances; upload Final Quotation, Closure Agreement & Chemical Spec Sheet.
Order Created	Platform / Agent	Generate Order ID; Proforma Invoice if advance payment; reserve inventory or schedule production; 24h cooling-off.
Confirmed	Buyer + Supplier (Ops finalizes)	Confirm details; lock ship date; move inventory reserved → committed; issue Order Confirmation Note.
Processing	Supplier (Warehouse / Production)	Blend / repackage; QC testing; apply GHS & UN marks; upload CoA, Batch Report & draft Packing List.

Status	Responsible Party	Required Action (key)
Shipping	Supplier / Logistics Partner	Record carrier, B/L & seal nos.; generate final Commercial Invoice; upload B/L–AWB, Packing List, MSDS + DGD.
In Transit	Logistics Partner (auto-updates)	Poll carrier API; display location & ETA to buyer; flag route deviations or delays immediately.
Customs / Clearance	Agent / Logistics (Customs Broker)	Upload customs declaration package; cycle Submitted → Awaiting Release → Cleared; pay duties.
Exception: Customs Hold	Agent (Compliance) → Admin	Correct & re-upload rejected docs; resubmit; update buyer every 24h; auto-assign lead after 5 days.
Delivered	Buyer / Logistics Partner	Confirm receipt; upload POD; record timestamp + GPS; open 7-day quality acceptance window.
Completed	Platform / Agent (or System)	Verify payment Settled / LC released; archive documents; order becomes read-only.



CHAPTER 5:

ROLE PERMISSION MATRIX



Buyer

Confirms order, monitors execution, receives goods, manages payment.



Supplier

Prepares goods, provides documents, hands off to logistics.



Agent

Coordinates operations, ensures compliance, manages exceptions & disputes.



Logistics Partner

Manages physical movement from origin to destination.



Platform / Admin

Governs system behavior, audit integrity & dispute resolution.



SYSTEM ACTIONS, DOCUMENTS & AUTHORITY

Role	System Actions	Document Actions	Transition Authority	Escalation
Buyer	View own orders, payment status & docs · Create RFQ acceptance, PO, tickets, disputes · Update delivery address (pre-Shipping) · Confirm terms, receipt, resolution.	Upload PO & POD · View all order docs · Cannot delete, modify or reject any document.	Order Created → Confirmed; Delivered → Completed	Flags to assigned Agent (can not escalate directly)
Supplier	View assigned orders only (by supplier_id) · Create uploads, availability & inability-to-fulfill flag · Update processing % · Confirm dispatch.	Upload (gated): CoA, Batch Report, Packing List, B/L, Commercial Invoice, MSDS, DGD.	Confirmed → Processing (batch ID); Processing → Shipping	Inability-to-fulfill → Agent (24h SLA)
Agent	View ALL active orders, exception queues, SLA timers, audit trails · Create assignments, escalations, compliance cases · Update exception status & SLA extensions.	Validate all docs; request resubmission with feedback · Can not modify document content.	Can force any transition with full audit trail	Level 1 for confirmations & docs → Platform Admin

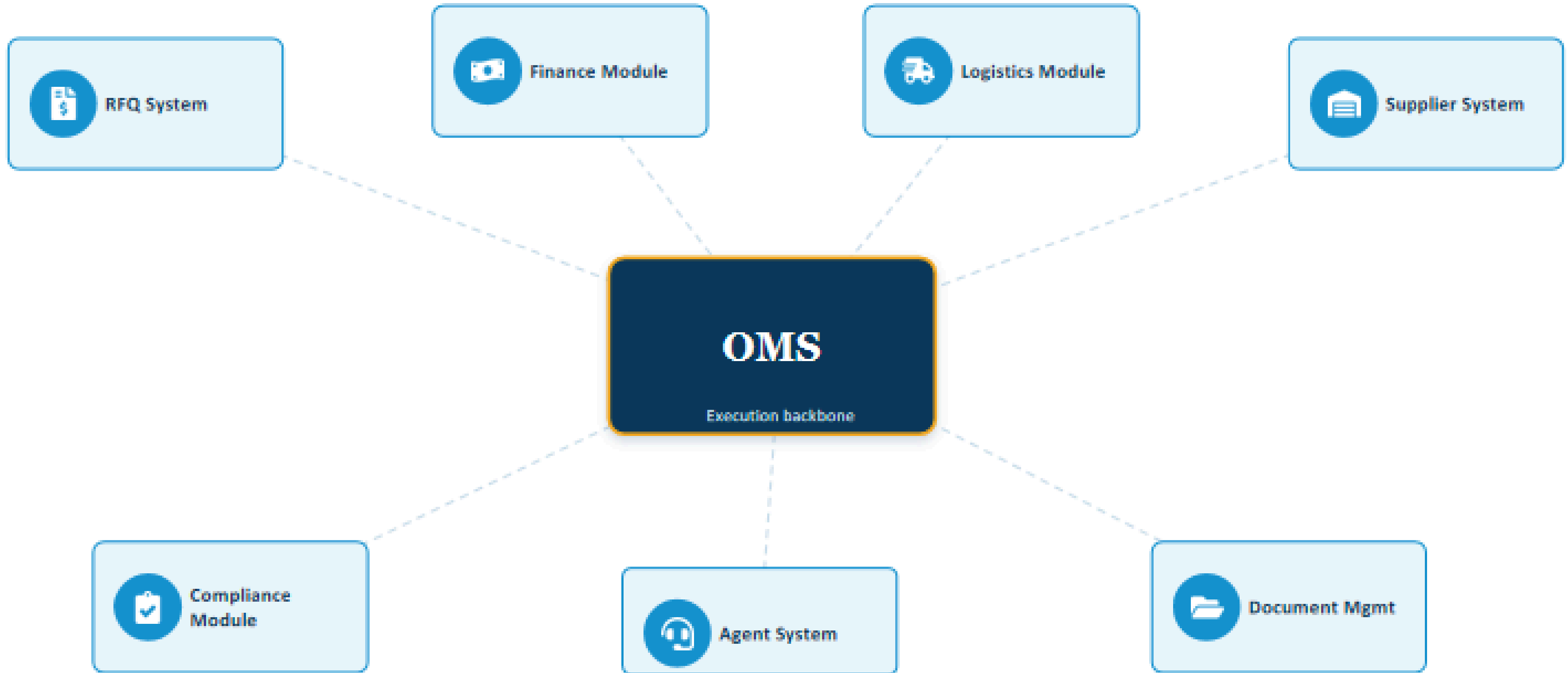


SYSTEM ACTIONS, DOCUMENTS & AUTHORITY

Role	System Actions	Document Actions	Transition Authority	Escalation
Logistics Partner	View logistics fields only (shipper, consignee, weight, hazmat, destination) · Create checkpoint & POD uploads · Up date real-time tracking & delay flags · Cannot view pricing.	Upload B/L copy, tracking proof, POD, customs release docs · Read-only on all other docs.	In Transit → Delivered (via valid POD)	Delay → Logistics Manager → Platform Admin (24h)
Platform / Admin	View ALL data + audit logs + escrow · Create roles & override rules · Update any field, workflow, SLA timers · Delete archived orders only (post 2-yr).	Approve resubmissions, extend SLA, force document acceptance (audited), archive orders.	Force any transition; every override logged (reason, time, admin ID)	Final point; Dispute → Senior Management (48h)



CHAPTER 6: INTEGRATION MAP



BIDIRECTIONAL DATA FLOWS & SLAS

System	Data → OMS	OMS → System	Error Handling & SLA	System
RFQ System	Product specs (SKU, UN no., purity), volume, destination, buyer identity, quoted terms, preferred Incoterms.	Conversion status with Order ID; completion + fulfillment date; shipped-vs-ordered variance %.	Order proceeds if down (retry 5 min/1h) ; invalid SKU blocks creation. Ack within 5s.	RFQ System
Finance Module	Advance-pay flag, payment terms, escrow ID, milestones, invoice status, currency & FX, tax codes.	Stage transitions triggering milestones; shipment value; POD; dispute flag freezing payment.	Payment 'pending' if down — does NOT auto-complete; manual Admin step. Escrow within 2h.	Finance Module
Logistics Module	Freight options & pricing, mode & route, transit estimate, real-time tracking, customs status, POD.	Shipment details for quoting; booking trigger at Confirmed; cancellation & delay-escalation signals.	Shipping blocked if down, Agent alerted Tracking stale >48h → escalate. API <3s, retry 2×.	Logistics Module
Supplier System	Real-time inventory, production readines, fulfillment date, batch/lot + shelf-life, QC status.	Order confirmation signal, batch-assignment request, doc reminders, cancellation, pickup confirmation.	Order stuck in Confirmed if down. Qty conflict → 24h to resolve. QC failure → Quality Hold.	Supplier System



BIDIRECTIONAL DATA FLOWS & SLAS

System	Data → OMS	OMS → System	Error Handling & SLA
Compliance Module	Required-doc checklist per product & destination, HS/tariff codes, DG classification, duty %, validity periods.	Product + destination for checks; document submissions for validation; pre-customs notification.	Order flagged 'Compliance Pending' if down — cannot enter Customs. Checks <10s; escalate after 2 rejections.
Agent System	Coordinator identity & contact, escalation ownership, SLA timers, comm-thread refs, workload metrics.	Exception triggers with deadline & priority; status changes; new-message & SLA-escalation signals.	Orders still process if down; assignment fails → Admin manual. No ack in 1h → Agent Manager.
Document Management	Upload status, storage URL, version history, metadata (issue, authority, expiry), integrity hash.	Store docs per stage; archive full package at Completed; retrieve for audit & buyer download.	Uploads & gated transitions blocked if down. Upload <30s incl. virus scan; hash mismatch → reject.

INTEGRATION PRINCIPLES

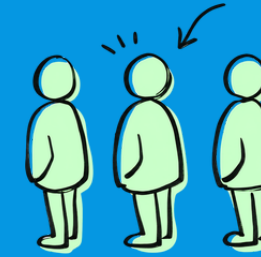
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|---|
| 1. Graceful degradation — blocked transitions are flagged |
| 2. Async for non-critical; sync for payment, inventory & compliance |
| 3. Every call logged for audit (payload, response, latency) |
| 4. Retry with backoff; alert Admin after max retries |
| 5. Incoming data validated against schema before any state update |



CHAPTER 7: COMMUNICATION PLAN

COMMUNICATION EVENT MATRIX

9 Critical Events Across the Order Lifecycle | TradeChem OMS



No	Trigger Event	Recipient(s)	Message Purpose	Channel
1	Order Created	Buyer, Supplier, Agent	Confirm order generation, share Order ID, outline next actions	Dashboard + Email + WhatsApp
2	Supplier Confirmation	Supplier	Request supplier to confirm availability and terms before deadline	Dashboard + Email
3	Document Upload Required	Supplier, Agent	Notify required document not uploaded; must submit to advance	Dashboard + Email
4	Order Confirmed by All	Buyer, Supplier, Agent	Confirm all parties committed; order in active processing	Dashboard + Email
5	Shipment Dispatched	Buyer, Agent	Inform goods handed to logistics; provide tracking and ETA	Dashboard + Email
6	Delay Detected	Buyer, Agent, Logistics	Alert shipment behind schedule; include new ETA and action	Dashboard + Email + WhatsApp
7	Customs Hold Detected	Agent, Compliance, Logistics, Buyer	Explain hold reason; assign compliance team; provide correction link	Dashboard + Email + WhatsApp
8	Delivery Confirmation	Buyer	Request buyer confirm receipt and report discrepancies	Dashboard + Email
9	Order Completed	Buyer, Supplier, Agent	Confirm order fully closed, obligations fulfilled, payment released	Dashboard + Email

MESSAGE STRUCTURE & CHANNEL STRATEGY

MESSAGE STRUCTURE

Required components for every system-generated message



ORDER ID

Unique identifier assigned at order creation, referenced in all communications prominently displayed



TIMESTAMP

Exact date and time of the triggering event in local timezone for auditable SLA tracking



ACTION LINK

Direct URL to relevant system function for immediate one-click response and action



ESCALATION CONTACT

Contact info for next-level responsible party if SLA expires without action taken



MESSAGE PURPOSE

Clear statement of why the notification is sent and what response is expected from recipient

CHANNEL STRATEGY

Tiered multi-channel approach based on event priority



DASHBOARD ALERT

ALL EVENTS

Persistent, context-rich alerts within the TradeChem platform. Serves as the primary record of all communications. Always active and accessible.



EMAIL NOTIFICATION

ACTION / DOCUMENT

Detailed, formal notifications with embedded action links. Suitable for events requiring documentation or formal acknowledgment and archival.



WHATSAPP BUSINESS

URGENT ONLY

Immediate, high-visibility alerts for time-sensitive events. Dominant in Indonesia for business communication. Mandatory for high-priority events.

• All communications must include Order ID, Timestamp, Action Link, and Escalation Contact



ESCALATION PROTOCOL & SLA FRAMEWORK

Automatic escalation chain when SLA windows expire without action

Event Type	SLA Window	Escalation Level 1	Escalation Level 2
Supplier Confirmation	24 Hours	Assigned Agent	Platform Admin
Document Upload	48 Hours	Assigned Agent	Platform Admin
Delivery Confirmation	72 Hours	Assigned Agent	Platform Admin
Delay Resolution	24 Hours	Logistics Manager	Platform Admin
Dispute Response	48 Hours	Platform Admin	Senior Management



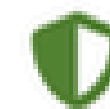
AUTOMATIC TIMER

Escalation timer begins at the moment the initial notification is sent. System automatically triggers escalation if action is not completed within the SLA window.



ESCALATION CONTENT

Each escalation includes the original notification details, elapsed time, and the specific action required from the escalation recipient.



ADMIN OVERRIDE

Platform Admin has override authority to extend SLA windows, reassign responsibilities, or force status transitions with all actions logged in audit trail.



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THANK YOU